

INTRODUCTION TO CYBER SECURITY

1. Meaning of Cyber Security

Cyber Security is the practice of **protecting computers, networks, systems, and data** from **digital attacks**, unauthorized access, damage, or theft. These attacks are often called **cyber threats** or **cyber attacks**.

2. Importance of Cyber Security

Cyber security is important because:

- Protects **personal data** (passwords, photos, bank details)
 - Protects **organizational data** (hospital records, school systems, government data)
 - Prevents **financial loss**
 - Ensures **privacy and trust**
 - Supports safe use of the **internet and digital services**
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3. Common Cyber Threats

Some common cyber threats include:

- **Malware** – harmful software such as viruses, worms, and trojans
 - **Phishing** – fake emails or messages tricking users to give personal information
 - **Ransomware** – locks files and demands payment
 - **Hacking** – unauthorized access to systems
 - **Spyware** – secretly monitors user activity
 - **Denial of Service (DoS)** – makes a system unavailable to users
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4. Key Areas of Cyber Security

Cyber security focuses on protecting:

- **Network Security** – protects networks from attacks
- **Application Security** – secures software and applications
- **Information Security** – protects data from unauthorized access
- **Operational Security** – manages permissions and access control
- **End-User Security** – educating users on safe practices

5. Cyber Security Principles (CIA Triad)

Cyber security is based on three main principles:

1. **Confidentiality** – only authorized people access data
2. **Integrity** – data is accurate and not altered
3. **Availability** – systems and data are accessible when needed

6. Basic Cyber Safety Practices

- Use **strong passwords**
- Do not share personal information online
- Avoid clicking suspicious links
- Keep software updated
- Use antivirus and firewalls
- Log out from public computers

7. Careers in Cyber Security

Cyber security offers careers such as:

- Cyber Security Analyst
- Ethical Hacker
- Network Security Engineer
- Digital Forensics Expert

8. Conclusion

Cyber security is essential in today's digital world. As technology grows, cyber threats also increase, making cyber security a **shared responsibility** for individuals, organizations, and governments.